DOD-D-24577(SH) METRIC
18 September 1978

MILITARY SPECIFICATION

DISTILLER SCALE PREVENTIVE TREATMENT FORMULATIONS (METRIC)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general purpose distiller scale preventive treatment formulations.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect in date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
NN-P-71 - Pallet, Material Handling, Wood, Stringer Construction, 2 Way And 4 Way (Partial).

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking For Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

FEDERAL
FED-STD-595 - Color.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D1141 - Substitute Ocean Water.
D1293 - pH of Water and Waste Water.
D1426 - Ammonia Nitrogen in Water.
D1881 - Foaming Tendencies of Engine Coolants in Glassware.
D2576 - Metals in Water and Waste Water by Atomic Absorption Spectrophotometry.
D2688 - Corrosivity of Water in the Absense of Heat Transfer (Weight Loss Methods).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

DEPARTMENT OF TRANSPORTATION (DOT)
Code of Federal Regulations - Title 49, Part 178.19.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4620

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INCORPORATED, AGENT
National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Uniform Classification Committee Agent, G. F. Earl, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The distiller scale preventive treatment formulations furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List (QPL) at the time set for opening of bids (See 4.3 and 6.3).

3.2 Material. The distiller scale preventive treatment formulations shall contain organic polyelectrolytes such as polyacrylates, and antifoam agent in aqueous solution which shall conform to the requirements specified hereinafter.

3.2.1 Unit content. The unit content shall be 6.0 gallons (or Metric equivalent) of liquid at 25°C (77°F) as specified in 4.5.1.

3.2.2 Maintenance of heat transfer coefficient. Using the recommended dosage of the distiller scale preventive formulation in the feedwater at a concentration of 10.5 + 0.2 milligram per liter (mg/L) the heat transfer coefficient shall be maintained at no less than 80 percent of the "clean tube value" throughout the 80-hour test period when tested in accordance with 4.6.1.

3.2.3 Scale removal. Scale formed on heat transfer surfaces, while using the distiller scale preventive treatment formulation, shall be removed by one acid cleaning using the procedure given in 4.6.1.2.

3.2.4 Appearance. The distiller scale preventive treatment formulation shall be a clear liquid with no phase separation or shall form a stable emulsion when examined by the procedure given in 4.6.2.

3.2.5 Toxicity. The distiller scale preventive treatment formulation shall not contain toxic materials in concentrations which would constitute a health hazard. The formulation shall be safe to use for production of potable water by distillation (see 4.6.3).

3.2.6 Corrosion. The distiller scale preventive treatment formulation shall not corrode 90-10 copper-nickel (Cu-Ni), a distiller heat transfer metal, at a rate greater than 2.54×10^{-2} millimeters per year (mm/yr) (1 (mil/yr)) when 90-10 Cu-Ni metal is immersed in 100 milligram per liter (mg/L) concentration of distiller scale preventive treatment formulation in distiller water. The corrosion rate shall not be exceeded when tested in accordance with 4.6.4.

3.2.7 Compatibility with other treatment formulations. Distiller scale preventive treatment formulations shall be compatible with formulations qualified under this specification. Compatibility shall be defined as the absence of a precipitate when the distiller candidate treatment formulation is mixed in the concentration recommended for use in the distiller chemical addition tank with each formulation. The testing procedure for this requirement is given in 4.6.5.

3.3 Chemical properties.

3.3.1 Ammonia content. The ammonia content of the scale preventive treatment shall not exceed 110 mg/L when analyzed in accordance with 4.7.1.

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3.3.2 Hydrogen ion concentration (pH). The value for qualification as measured at 25°C ± 1°C shall be not less than 6.5 or not more than 10.5. The pH of subsquent production lot samples shall vary not more than 0.8 pH units from the value obtained for the qualification sample.

3.3.3 The iron content. The iron content of the scale preventive treatment shall not exceed 10 mg/L in accordance with 4.7.3.

3.3.4 Polyelectrolyte content. The scale preventive treatment shall contain at least 14.0 percent by weight of the polyelectrolyte chemical when tested in accordance with 4.7.4. The polyelectrolyte concentration shall not vary more than 0.55 weight percent from the value obtained for qualification testing.

3.4 Physical properties.

3.4.1 Antifoaming action. The distiller treatment formulation, shall control foaming to prevent brine carryover into the distillate. Antifoaming action shall meet the test requirements of 4.8.1.

3.4.2 Density. The density of the liquid shall not vary more than 0.01 grams per milliliter (g/mL) from the value obtained on qualification scale preventive. Density test shall be in accordance with 4.8.2.

3.4.3 Intrinsic viscosity. The intrinsic viscosity shall be determined at 25.0°C ± 0.1°C by the dilution method using a Cannon-Ubbelohde viscosimeter. The intrinsic viscosity shall be in accordance with 4.8.3. The intrinsic viscosity of subsequent lots shall not vary more than 0.004 liter per gram (L/g) from the value obtained when the distiller scale preventive treatment chemical is determined for the qualification sample.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection of the distiller scale preventive treatment formulations shall be classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

4.3 Qualification inspection. Qualification shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center (NAVSEC). Qualification inspection shall consist of the examinations and test shown in table I.

4.3.1 Sample for qualification tests. One filled 6.0 gallon (or Metric equivalent) container is required for the qualification tests specified in table I.

TABLE I. Categories of inspection.

Title	Qualification	Quality Comformance	Requirement paragraph	Inspection paragraph
Examination of filled containers	x	x	3.2.1	4.5.1
Maintenance of heat transfer coefficient	x	---	3.2.2	4.6.1 and 4.6.1.1
Scale removal	x	---	3.2.3	4.6.1.2
Appearance	x	x	3.2.4	4.6.2
Toxicity	x	---	3.2.5	4.6.3
Corrosion	x	---	3.2.6	4.6.4

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TABLE I. Categories of inspection. - Continued

Title	Qualification	Quality Conformance	Requirement paragraph	Inspection paragraph
Compatibility of other treatment formulations	x	---	3.2.7	4.6.5
Ammonia content	x	x	3.3.1	4.7.1
Hydrogen ion concentration (pH)	x	x	3.3.2	4.7.2
Iron content	x	x	3.3.3	4.7.3
Polyelectrolyte content	x	x	3.3.4	4.7.4
Antifoaming action	x	x	3.4.1	4.8.1
Density	x	x	3.4.2	4.8.2
Intrinsic viscosity	x	x	3.4.3	4.8.3

4.4 Sampling for quality conformance inspection.

4.4.1 Inspection lot. For purposes of sampling, a lot shall consist of all material manufactured as one batch and transferred from one mixing tank to the shipping container.

4.4.2 Sampling for examination of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I. The acceptable quality level = 2.5 percent defective to verify compliance with all requirements regarding fill, closure, marking, and other requirements not requiring tests as specified in 4.5.1.

4.4.3 Sampling for tests. Two one-liter containers shall be filled with the distiller scale preventive formulation which shall be withdrawn from an agitated mixing tank prior to packing each lot. The samples shall be contained in plastic bottles such as polyethylene or other plastic containers which do not react with the chemical of the distiller scale preventive treatment. The three equal portions of each sample shall be used as follows:

(a) One for quality conformance tests (see 4.4).
(b) One for contractor.

The results of the tests required by 4.5.2 shall be sumitted to NAVSEC or the designated laboratory.

4.5 Quality conformance inspection.

4.5.1 Examination of filled containers. Each sample-filled container selected shall be examined for defects of the containers and the closure, for evidence of leakage, and for unsatisfactory markings. Each sample-filled container selected shall be measured to determine the volume of the contents. To determine conformance with 3.2.1, the liquid, maintained at 25.0°C ± 1°C (77°F ± 1.8°F) shall be measured in a precalibrated 6.0 gallon vessel (or Metric equivalent). Any container in the sample having one or more defects, or under required fill shall be cause for rejection, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, this shall be cause for rejection of the lot represented by the sample.

4.5.2 Quality conformance tests. The samples selected in accordance with 4.4.3 shall be subjected to the tests of table I, as applicable.

4.5.2.1 Action in case of failure. If the sample tested is found to be not in conformance with any requirement of this specification, the lot represented by the sample shall be rejected.

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4.6 Test procedures for material properties.

4.6.1 Maintenance of heat-transfer coefficient. The heat-transfer coefficient, (U), shall be determined by using a calibrated model submerged tube distiller such as the 38 liters per hour (L/h) (10 gallons per hour (gal/h)) unit schematically shown on figure 1. The heat-transfer coefficient after 10 hours of operation is designated as the "clean tube value." The "clean tube value" shall be greater than 2555 watts per meter^2 kelvin, 450 British thermal unit per hour - square foot degree F BTU/(ft^2.h.°F). Test periods of distiller operation are conducted with synthetic seawater conforming to ASTM D1141. The seawater is prepared by mixing sea salts in the proportion of 41 grams of salt per liter of water (5.5 ounces of salt per gallon), resulting in a 35,000 mg/L solution of sea salts. The distiller scale preventive treatment is then added to the seawater in a concentration of 10.5 ± 0.2 mg/L. The temperature data needed to calculate the heat-transfer coefficient are recorded every two hours during periods of operation. The distiller is operated at 85°C, (185°F), boiling temperature, and the average feedwater flow rate is 0.45 gallon per minute (gal/min). The steam supplied by the boiler is at an average temperature of 108°C (226°F) at a gage pressure of 19.1 lb/in^2. The brine concentration ratio is maintained at 1.5 ± 0.1 and the feedwater (seawater) is maintained at pH 8.2 ± 0.2. The test is conducted for 90 hours including an initial 10-hour break-in period.

4.6.1.1 Heat-transfer coefficient calculations. The heat-transfer coefficient, "U", is expressed by the equation

$$U = Q/A\Delta T$$

ΔT (in units of °C or °F) is the apparent temperature difference between the saturated steam temperature and the boiling temperature in the evaporator chamber. "A" is the heat-transfer surface area in units of square meters (m^2) or square feet (ft^2), and Q is the heat flow rate in watts (W) or BTU/hr. "Q" is calculated by knowing the feedwater flow rate and by measuring the temperature prior to and immediately after passage through the heat exchanger. The heat-transfer coefficient U is affected by scaling of heat-transfer surfaces during operation of the distiller. After the first 10 hours of the test run, the heat-transfer coefficient is calculated by using the value of A for the distiller and the measured value of ΔT. The heat-transfer coefficient is calculated for every 2 hours of operation.

$$\text{percent "Clean tube value"} = \frac{U_2}{U_1} \times 100$$

U_1 = Clean tube heat-transfer coefficient BTU/ft^2.h.°F after 10 hours of operation.

U_2 = Heat-transfer coefficient BTU/ft^2.h.°F determined at 2-hour intervals throughout 90 hours of operation.

4.6.1.2 Scale removal. At the completion of the 90-hour test period, the ease of scale removal is determined by cleaning the distiller with 0.90 kilograms (two pounds) of uninhibited sulfamic acid, dissolved in 18.1 kilograms (40 pounds) of water, making a 5 percent weight solution. This solution is injected into the distiller and allowed to stand for one hour; the distiller is heated to 93.3°C (200°F) for another hour, and then rinsed. Using the same test conditions that were used during the 90-hour test run, the heat-transfer coefficient for the distiller heat transfer surfaces shall be at least as high as the value obtained previously at zero hours.

4.6.2 Appearance. The distiller scale preventive treatment formulation shall be visually examined. It shall be a clear liquid with no phase separation, within a temperature range of 25°C (77°F), to 60°C (140°F).

4.6.3 Environment and health compatibility. The distiller scale preventive treatment formulation shall be certified by Bureau of Medicine and Surgery (BUMED) to be safe for use to produce potable water by seawater distillation.

4.6.4 Corrosion. A 90-10 Cu-Ni coupon shall be placed in a one-liter Erlenmeyer flask and 500 mL of distilled water containing 100 mg/L or distiller scale preventive treatment formulation shall be added. A water-cooled condenser shall be attached to the flask. The solution is heated on an electric hot plate and allowed to reflux for 30 hours. The coupon preparation and calculation of corrosion rate shall be conducted according to procedure B of ASTM D2688.

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4.6.5 Compatibility with other treatment formulations. One hundred mL of a 10 percent solution of the distiller scale preventive treatment formulation shall be mixed at 25°C ± 1°C (77°F ± 2°F) with 100 mL of each solution previously qualified. The material being tested is not compatible if a percipitate forms when the two solutions are mixed.

4.7 Test procedures for chemical properties.

4.7.1 Ammonia content. No greater than 110 mg/L of ammonia shall be present in a sample of the liquid. The analysis shall be made according to ASTM D1426 by the distillation method using the Nessler reagent.

4.7.2 pH value. The pH of the liquid shall be greater than 6.5 and less than 10.5. The pH shall be within 0.8 pH unit of the value obtained as specified in 3.3.2. The method for this measurement is described in ASTM D1293.

4.7.3 Iron content. The iron concentration of a sample shall be determined by atomic absorption analysis as described in ASTM D2576 and shall not exceed the iron content of 3.3.3.

4.7.4 Polyelectrolyte content - acetone-insoluble material. The solution used to determine the density (see 4.8.2), shall be used for this determination. Twenty-five mL of reagent grade acetone (water, 0.5 percent maximum) is added to the sample and a small stirring rod is used to bring the acetone into contact with the lower layer. After 2 or 3 minutes of stirring the mixture, the tube is centrifuged for 5 minutes at a speed by which the centrifuge is capable of producing a relative centrifugal force (rcf) between 600 and 700 at the tips of the tubes. The slightly milkly supernatant liquid layer is removed from the gummy polymer residue. A second 25 mL portion of acetone is added to the residue and again brought into contact with the insoluble material with a stirring rod. The stirring is continued until the acetone becomes turbid and a crystalline-like material forms. The tube is capped, and centrifuged again for 5 minutes at 1500 revolutions per minute (r/min) and the supernatant acetone extract is again removed from the solid residue. The acetone treatment is repeated with a third 25 mL portion of acetone. The solid polymer material remaining in the 50 mL centrifuge tube is gently warmed on a hot plate (surface temperature 60°C (140°F) in a well-ventilated hood. The solid is then dried in an oven set at 105°C ± 2°C (221°F ± 3.6°F) for 1 hour, cooled in a desiccator over a suitable desiccant and weighed with the tubes capped (the dried solid may be hygroscopic). The drying is repeated until successive weighings are within ± 1 mg. The formulation will not be acceptable if the solid residue is oily or gummy. The percentage of acetone insoluble polyelectrolyte shall be within ± 0.55 weight percent of the value obtained as specified in 3.3.4.

$$\text{percent acetone insoluble material} = \frac{\text{wt of material in 50 mL centrifuge tube x 100}}{\text{wt of original liquid sample}}$$

4.8 Test procedures for physical properties.

4.8.1 Antifoaming action. The test is a modification of ASTM D1881. A gas dispersion tube is used which is 250 mm long and has a 12 mm outside diameter coarse fritted glass cylinder. A flowmeter is placed between a tank of breathing-quality compressed air and the bubbler. A flow of 1 L/min is regulated by the pressure valve on the tank. A vigorously shaken 250 mL sample of liquid formulation and 1 molar sodium chloride solution are used to prepare a 1 percent solution that is 0.2 M with respect to sodium chloride. The temperature of the test shall be 90°C-95°C (194°F-203°F). With these modifications, the ASTM procedure is then followed. No foam should persist immediately after the air is secured; that is, the "eye" should form immediately.

4.8.2 Density. A 2 mL portion of the well mixed liquid formulation at 25°C + 1°C (77°F + 1.8°F) shall be measured by a volumetric pipette and transferred to a dried, preweighed 50 mL centrifuge tube with a cap. The liquid shall be weighed (± 1 mg) with the cap on the tube to prevent errors due to evaporation.

$$\text{Density} = \frac{\text{weight of liquid (g)}}{\text{volume of liquid (mL)}}$$

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4.8.3 Intrinsic viscosity. The intrinsic viscosity shall be as follows:

(a) The intrinsic viscosity shall be determined at 25°C ± 0.1°C (77°F ± 0.2°F) by the dilution method using a Cannon-Ubbelohde viscosimeter with a shear range at the wall of 45 to 1800 sec^{-1}. The first step is to determine the reduced specific viscosity. For the necessary calculations of the reduced specific viscosity, a concentration of 100 grams per liter (g/L) for the active ingredients shall be assumed for the original liquid formulations. Three samples of the well mixed liquid, 10 mL (20 g/L), 7.5 mL (15 g/L) and 5 mL (10 g/L) are pipetted into separate 50 mL volumetric flasks. Add 10 mL of 1 M sodium chloride solution. Bring to the mark with distilled water to produce a final 0.2 M sodium chloride solution.

(b) The reduced specific viscosity ($\bar{\eta}$) expressed in L/g, is calculated from the time (t_o) obtained with the 0.2 M sodium chloride solution, and times (t_c) obtained with the diluted samples.

$$\bar{\eta} = \frac{t_c/t_o - 1}{C}$$

Where C is the concentration of active ingredients in g/L. The shear rates at the wall ($\dot{y}$) of the viscosimeter expressed in reciprocal (sec^{-1}), are determined by using the calibration constants (k) of the viscosimeter and the times, t_c.

A plot of $\bar{\eta}$ versus $\dot{y}$ is constructed and the shear rate at 1000 sec^{-1} is determined. Finally, the intrinsic viscosity (L/g) is calculated as the y-intercept of a least-squares fit of the reduced specific viscosities at a shear rate at 1000 sec^{-1} versus the concentration.

4.9 Inspection of preparation for delivery. Preservation-packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation-packaging. Preservation-packaging for levels A and C shall be as specified herein.

5.1.1 The distiller scale preventative formulation shall be furnished in a 6.0 gallon, plastic container as specified herein.

5.1.1.1 Plastic container. The container shall be molded polyethylene meeting the requirements of Department of Transportation (DOT) specification Number 34 as specified in the Code of Federal Regulations, Title 49, Part 178.19 and as follows:

(a) The container, closure, liner or space fillers shall not interact physically or chemically with the contents of the container so as to be altered or to alter the strength, quality, or purity of the container contents for at least four years from the date or manufacture.
(b) Shall be stackable and self-supporting.
(c) Shall be provided with a self venting flexible spout type closure.
(d) Shall be provided with an integrally moded or recessible plastic or metal handle. Metal handles shall not exceed a magnetic permeability of 2.0.
(e) Shall be a white color approximating color number 17875 of FED-STD-595.

5.2 Packing. Packing for levels A, B and C shall be as specified hereinafter.

5.2.1 Palletization. Twenty-four 6.0 gallon, (or Metric equivalent) plastic containers shall be palletized in accordance with the requirements of MIL-STD-147, Load Type XVII. Pallets conforming to NN-P-71, Type V class, wood group optional, size 2 are acceptable. Containers shall be properly and firmly nested and arranged to insure a smug, non-shifting load. Pallet dimensions may be adjusted to assure a snug, non-shifting load but shall not exceed 170 x 205 mm.

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5.2.2 Method of shipment shall comply with Uniform Freight or National Motor Freight Classification Rules or regulations or other carrier rules as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required herein (see 6.2), unit containers and palletized unit loads shall be marked in accordance with MIL-STD-129. Unit container markings shall include:

(a) Manufacturer's or contractor's name or both
(b) Lot number
(c) Contract or order number
(d) Date of manufacture (month and year)
(e) Precautionary labeling
(f) Shelf life (expiration date of 48 months from date of manufacture, for example manufactured 1/75, expires 1/79)
(g) "MANDATORY SHELF LIFE MATERIAL, DISCARD MATERIAL AFTER EXPIRATION DATE" ( ).
(h) Do not store in direct sunlight.
(i) For transfer at sea operations, other than full pallet loads, breakout units should be places on the top of the transfer load.

6. NOTES

6.1 Intended use. The distiller scale preventive treatment chemical covered by this specification is for scale prevention for submerged tube, basket, and flash-type naval distillers. The distillate for distillers using this treatment shall be used as potable water and makeup feedwater for all Navy boilers. A dosage rate of one pint of scale preventive treatment per 4000 gallons of dissilate is standard for Navy shipboard use.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Special markings (see 5.3)

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL-24577 whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 4620-N029)

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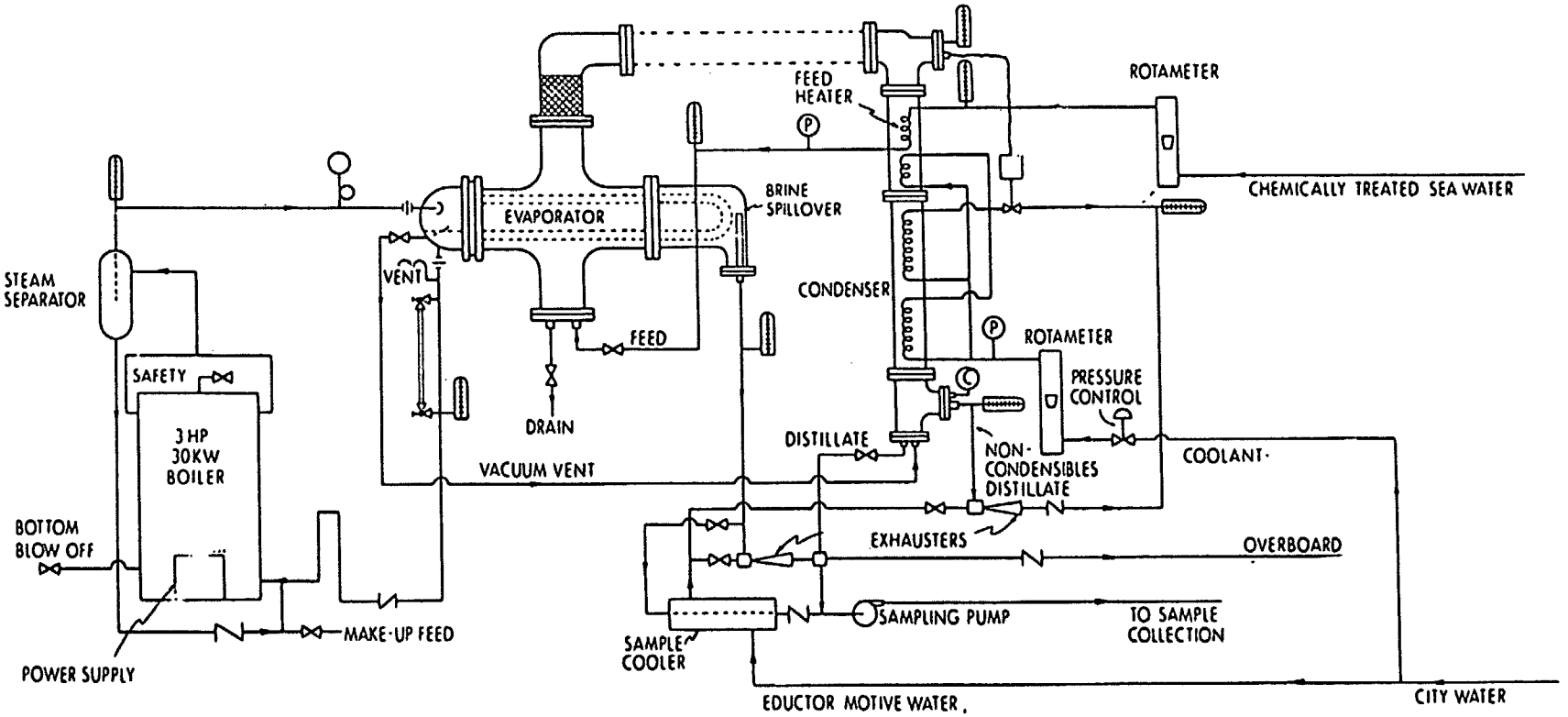


FIGURE 1. Diagrammatic arrangement-laboratory 38 L/h (10 gal/h) distillation system.

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